

# Start here: C2

Access and asset requests fulfilled against a role-based entitlement policy

## THE STORY

Ashvaka Industries Group runs construction sites, cement plants, a solar park and a warehouse across Gujarat and Maharashtra, and its IT service desk in Ahmedabad gets about a dozen requests a day for software access, licences, laptops, phones and site tablets. Each request has to be checked against a matrix that says which access each job role may hold, at what level, for how long, and with whose approval. People change jobs all the time, and the identity system often learns about a move days after HR does, so the directory can show a role the person has already left or not yet started. On top of that, licences run out, approvals arrive from people with no authority to give them, and temporary access stays switched on long after the need has passed. The AI Employee works out who is asking and what role they held on the day, grants what that role allows, sends everything else to a named approver with the gap spelled out, and puts an end date on every temporary grant.

## OPEN THESE THREE FILES FIRST

- dataset/reference/Access\_Policy.md** The rules the decision follows; read sections 3, 5, 6, 10 and 16 first.
- dataset/csv/Role\_Entitlement\_Matrix.csv** The core of the build: which entitlements each role may hold, the level, the time limit and the approval needed.
- dataset/csv/HR\_Role\_Changes.csv** Tells you the role a person actually held on the request date when sys\_user has not caught up.

## WORDS YOU WILL MEET

**RITM (request item)** ServiceNow's name for one requested item on a ticket, numbered like RITM0041628. It is the unit of work here: one person, one item, one decision, and every approval or email quotes it.

**Segregation of duties (SoD)** The rule that one person must not hold two powers that together allow fraud, such as creating a vendor and releasing payment to it. The forbidden pairs are listed in SoD\_Rules and every conflict goes to the CISO.

**Entitlement** One access right, licence or piece of hardware, identified by a code such as ACAD\_FULL or SAP\_MM\_PO\_CREATE. Requests often name it loosely, so the first job is to map what was typed to the right code.

**Licence pool** All the paid licences of one product, such as Power BI Pro, shared by everyone who needs it. When the pool is empty an in-role request waits in a queue for a freed or new licence; it is not refused.

**Role** The IAM role that comes with a job, such as SITE\_ENGINEER or AP\_CLERK. The role decides what a person may hold, so getting the role right on the day of the request decides almost everything else.

**Temporary grant and expiry** Access given until an end date, after which it must be removed. The owner gets a reminder three days before, and any extension needs a human decision.

**IAM (identity and access management)** The system and team that hold each person's account, role and grants and actually switch access on and off. Here IAM's view is sys\_user and Active\_Grants, and it can lag behind HR by up to ten days.

**Delegation** A dated arrangement where one approver lends their authority to another person, for example while on leave. It only counts between its start and end dates, inclusive.

**Birthright access** Access everyone in a role gets automatically, without anyone approving it, such as email, VPN and a standard laptop. It is marked Y in the birthright column of the matrix.

**Notice period** The time between someone resigning and their last working day. During it they get no new access, except handover access signed off by both their line manager and their company's HR business partner.

**Privileged access** Administrator or emergency rights, such as a firefighter ID, server admin or production database admin, that could do serious damage if misused. Only the CISO or Deputy CISO may approve it, and it is always temporary.

**Application owner** The named person accountable for an application, listed in Application\_Catalogue with a backup. They decide elevated and out-of-role access for that application.

#### ONE RECORD, END TO END

RITM0041628: AutoCAD for an engineer who has just moved out of design

- 1 In sc\_req\_item, RITM0041628 was raised by phone on 2026-10-01 11:30 IST by Hardik Raval (e37175) for himself, asking for AutoCAD (full) as Permanent, to mark up shop drawings.
- 2 Application\_Catalogue maps AutoCAD (full) to ACAD\_FULL in Autodesk AutoCAD: Standard level, licence pool LP-ACAD, owner Rutvik Joshi (E52098).
- 3 sys\_user still shows him as DESIGN\_ENGINEER, but HR\_Role\_Changes row HRC-26-1112 moved him to QA\_QC\_ENGINEER effective 29-09-2026, with recorded\_in\_iam\_on blank, so on 1 October his role was QA\_QC\_ENGINEER and the IAM record was stale.
- 4 Role\_Entitlement\_Matrix lists nothing in Autodesk AutoCAD for QA\_QC\_ENGINEER; ACAD\_FULL is carried only by DESIGN\_ENGINEER, so the request is out of role.
- 5 Out-of-role access goes to the application owner: Rutvik Joshi's delegation to Sameer Rao (AUTH-081) ended on 30-09-2026 and no approval is on the record, so it goes to Rutvik himself.
- 6 The AI Employee escalates to Rutvik Joshi with the gap (ACAD\_FULL is not in QA\_QC\_ENGINEER, only DESIGN\_ENGINEER carries it) and recommends, if approved, a temporary grant ending 31 Oct 2026 at the latest.
- 7 When the team releases staged reply 0006 from Rutvik Joshi, it says not approved, so the request is closed as declined.

Outcome. Escalate to Rutvik Joshi, the AutoCAD application owner, as out of role, stating that Hardik moved from Design Engineer to QA/QC Engineer on 29 September and that IAM has not caught up. No access is granted now; if Rutvik approved, the grant would be temporary to 31 Oct 2026 at most, never permanent.

Why. Policy section 3 says the role in force is set by the HR effective date, not by sys\_user, and section 6 sends access no role of the person carries to the named application owner with the gap, granting at most 30 days even if approved. A human decides because out-of-role access is always reserved for one (section 17).

#### WHAT THE ROUND 1 MINIMUM NEEDS

|                       |                                                                                                                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tables</b>         | access_request (sc_req_item), person (sys_user), hr_role_change (HR_Role_Changes), role_entitlement (Role_Entitlement_Matrix), application_catalogue (Application_Catalogue). Everything else in the pack is optional for the minimum, and every table is ready to import from setup_kit/ in this folder. |
| <b>Staged replies</b> | None. Staged reply 0006, the application owner's answer on the record above, is an optional extra.                                                                                                                                                                                                        |
| <b>Watch for</b>      | Judge the role on the request date, not today. Approvals sit in sys_approval and the mailbox, not on the request itself.                                                                                                                                                                                  |

#### WHAT THE JOB COSTS BY HAND

230 access and asset requests were raised between 21 Sep and 09 Oct 2026, about 360 a month. Time one by hand: minutes per record times records per month is what your AI Employee gives back.